Weikai Huang

Email: weikaih@cs.washington.edu | Homepage: https://weikaih04.github.io/ | Google Scholar

Short Biography

Weikai Huang is a PhD student at the Paul G. Allen School of Computer Science & Engineering, University of Washington, advised by Prof. Ranjay Krishna. He is also a student researcher at the Allen Institute for AI (AI2), and received his B.S. from UW in March 2026. His research lies in computer vision with an emphasis on 2D/3D grounding and tracking, controllable generative models, synthetic data generation for vision foundation models, and spatial/grounding-centric VLMs. His work has been published at top-tier venues including CVPR, ICLR, NeurIPS, and ECCV, and received multiple oral awards.

Education

- **University of Washington, Seattle** August 2026 – June 2031 (Expected)
  Ph.D. in Computer Science
  Advisor: Prof. Ranjay Krishna

- **University of Washington, Seattle** Sep 2023 – March 2026
  B.S. (with Honors) in Computer Science; GPA: 3.91
  Advisor: Prof. Ranjay Krishna

Selected Research (* indicates equal contribution)

- **MolmoAct2: Action Reasoning Models for Real-world Deployment**
  Haoquan Fang*, Jiafei Duan*, Donovan Clay, Sam Wang, Shuo Liu, **Weikai Huang**, Xiang Fan, Wei-Chuan Tsai, Shirui Chen, Yi Ru Wang, Shanli Xing, Jaemin Cho, Jae Sung Park, Ainaz Eftekhar, Peter Sushko, Karen Farley, Angad Wadhwa, Cole Harrison, Winson Han, Ying-Chun Lee, Eli VanderBilt, Rose Hendrix, Suveen Ellawela, Lucas Ngoo, Joyce Chai, Zhongzheng Ren, Ali Farhadi, Dieter Fox, Ranjay Krishna
  arXiv 2026
  Contribution: Contributed to the open vision-language-action system and data/model pipeline for real-world robot deployment.

- **WildDet3D: Scaling Promptable 3D Detection in the Wild**
  Weikai Huang, Jieyu Zhang, Sijun Li, Taoyang Jia, Jiafei Duan, Yunqian Cheng, Jaemin Cho, Matthew Wallingford, Rustin Soraki, Chris Dongjoo Kim, Shuo Liu, Donovan Clay, Taira Anderson, Winson Han, Ali Farhadi, Bharath Hariharan, Zhongzheng Ren, Ranjay Krishna
  arXiv 2026
  Contribution: Led the project. Built an open-vocabulary monocular 3D detection system supporting text, point, and box prompts. Curated WildDet3D-Data with 1M+ images and 3.9M annotations across 13K+ categories.

- **Imaginative Perception Tokens Enhance Spatial Reasoning in Multimodal Language Models**
  Mahtab Bigverdi*, Linjie Li*, **Weikai Huang***, Yiming Liu, Jaemin Cho, Jieyu Zhang, Tuhin Kundu, Chris Dongjoo Kim, Zelun Luo, Linda Shapiro, Ranjay Krishna
  ECCV 2026
  Contribution: Trained the omni model (BAGEL) and curated spatial reasoning data from Habitat and AI2-THOR for the omni model.

- **Synthetic Object Compositions for Scalable and Accurate Learning in Detection, Segmentation, and Grounding**
  Weikai Huang, Jieyu Zhang, Taoyang Jia, Chenhao Zheng, Ziqi Gao, Jae Sung Park, Winson Han, Ranjay Krishna
  CVPR 2026
  Contribution: Led the project. Trained 100+ model checkpoints with different configs using MMDetection and Detectron2. Processed 10M+ images using GCP and AI2 clusters.

- **Molmo2: Open Weights and Data for Vision-Language Models with Video Understanding and Grounding**
  AI2 Prior Team
  CVPR 2026 **(Best Paper Candidate; Compute Transparency Champion)**
  Contribution: Multi-image data curation and processing, model fine-tuning, benchmark setup and evaluation.

- **Generate Any Scene: Scene Graph Driven Data Synthesis for Visual Generation Training**
  Ziqi Gao*, **Weikai Huang***, Jieyu Zhang, Aniruddha Kembhavi, Ranjay Krishna
  ICLR 2026
  Contribution: Designed programmatic scene graph & prompt generation engine for evaluating and improving text-to-vision models. Conducted experiments on 30+ text-to-image, text-to-video, and text-to-3D models. Fine-tuned Stable Diffusion 1.5 with LoRA using DreamSync and DreamBooth methods.

- **Task Me Anything**
  Jieyu Zhang, **Weikai Huang***, Zixian Ma*, Oscar Michel, Dong He, Tanmay Gupta, Wei-Chiu Ma, Ali Farhadi, Aniruddha Kembhavi, Ranjay Krishna
  NeurIPS 2024
  Contribution: Designed programmatic benchmark generation engine capable of generating 750M+ VQA questions for multimodal language models. Implemented unified inference interfaces for 20+ MLMs and ran large-scale experiments on 100+ GPUs. Built 3D image/video generation and rendering pipeline with Blender.
- **m&m's: A Benchmark to Evaluate Tool-Use for Multi-Step Multi-Modal Tasks**
  Zixian Ma, **Weikai Huang**, Jieyu Zhang, Tanmay Gupta, Ranjay Krishna
  ECCV 2024
  Contribution: Implemented 20+ tool interfaces/APIs including image processing, segmentation, captioning, web search, and location search. Built human annotation pipeline and interface for high-quality data annotation.

Experience

- **Research Intern:** Salesforce AI Research, Palo Alto, CA — Jun 2026 – Sep 2026
- **Student Researcher:** Allen Institute for AI (AI2) — Jun 2025 – Jun 2026
- **Research Assistant:** UW CSE RAIVN Lab — Oct 2023 – Present

Services

- **Reviewer:** CVPR 2025
- **Organizer:** Synthetic Data for Computer Vision Workshop @ CVPR 2024, 2025, 2026
- **Teaching Assistant:** UW CSE 455 Computer Vision (Autumn 2025), UW CSE 493G Deep Learning (Spring 2025, Winter 2026)
- **Host:** 2024 UW CSE Education Panel

Award

- **Fellowships**
  - UW CSE John and JoAnne Wisniewski Endowed Scholarship (2 out of 2000 CS undergrads) — 2024
- **Academic Honors**
  - Outstanding Computer Science & Engineering Senior Award (one of three recipients selected from the graduating CSE senior class) — 2026
  - CVPR 2026 Compute Transparency Champion, Molmo2 — 2026
  - CVPR 2026 Outstanding Reviewer — 2026
  - UW Annual Dean's List — 2023, 2024, 2025, 2026

Publications (* indicates equal contribution)

- **MolmoAct2: Action Reasoning Models for Real-world Deployment**
  Haoquan Fang*, Jiafei Duan*, Donovan Clay, Sam Wang, Shuo Liu, **Weikai Huang**, Xiang Fan, Wei-Chuan Tsai, Shirui Chen, Yi Ru Wang, Shanli Xing, Jaemin Cho, Jae Sung Park, Ainaz Eftekhar, Peter Sushko, Karen Farley, Angad Wadhwa, Cole Harrison, Winson Han, Ying-Chun Lee, Eli VanderBilt, Rose Hendrix, Suveen Ellawela, Lucas Ngoo, Joyce Chai, Zhongzheng Ren, Ali Farhadi, Dieter Fox, Ranjay Krishna
  arXiv 2026
- **WildDet3D: Scaling Promptable 3D Detection in the Wild**
  Weikai Huang, Jieyu Zhang, Sijun Li, Taoyang Jia, Jiafei Duan, Yunqian Cheng, Jaemin Cho, Matthew Wallingford, Rustin Soraki, Chris Dongjoo Kim, Shuo Liu, Donovan Clay, Taira Anderson, Winson Han, Ali Farhadi, Bharath Hariharan, Zhongzheng Ren, Ranjay Krishna
  arXiv 2026
- **Imaginative Perception Tokens Enhance Spatial Reasoning in Multimodal Language Models**
  Mahtab Bigverdi*, Linjie Li*, **Weikai Huang***, Yiming Liu, Jaemin Cho, Jieyu Zhang, Tuhin Kundu, Chris Dongjoo Kim, Zelun Luo, Linda Shapiro, Ranjay Krishna
  ECCV 2026

- **Synthetic Visual Genome 2: Extracting Large-scale Spatio-Temporal Scene Graphs from Videos**
  Ziqi Gao, Jieyu Zhang, Wisdom Oluchi Ikezogwo, Jae Sung Park, Tario G. You, Daniel Ogbu, Chenhao Zheng, **Weikai Huang**, Yinuo Yang, Winson Han, Quan Kong, Rajat Saini, Ranjay Krishna
  ECCV 2026
- **Molmo2: Open Weights and Data for Vision-Language Models with Video Understanding and Grounding**
  Christopher Clark*, Jieyu Zhang*, Zixian Ma*, ..., **Weikai Huang**, ..., Ali Farhadi, Ranjay Krishna
  CVPR 2026 ***(Best Paper Candidate; Compute Transparency Champion)***
- **Synthetic Object Compositions for Scalable and Accurate Learning in Detection, Segmentation, and Grounding**
  Weikai Huang, Jieyu Zhang, Taoyang Jia, Chenhao Zheng, Ziqi Gao, Jae Sung Park, Winson Han, Ranjay Krishna
  CVPR 2026
- **TrajTok: Learning Trajectory Tokens Enhances Video Understanding**
  Chenhao Zheng, Jieyu Zhang, Jianing Zhang, **Weikai Huang**, Ashutosh Kumar, Quan Kong, Oncel Tuzel, Chun-Liang Li, Ranjay Krishna
  CVPR 2026
- **Generate Any Scene: Scene Graph Driven Data Synthesis for Visual Generation Training**
  Ziqi Gao*, **Weikai Huang***, Jieyu Zhang, Aniruddha Kembhavi, Ranjay Krishna
  ICLR 2026
- **ProVision: Programmatically Scaling Vision-centric Instruction Data for Multimodal Language Models**
  Jieyu Zhang, Le Xue, Linxin Song, Jun Wang, **Weikai Huang**, Manli Shu, An Yan, Zixian Ma, Juan Carlos Niebles, Silvio Savarese, Caiming Xiong, Zeyuan Chen, Ranjay Krishna, Ran Xu
  arXiv 2024
- **Task Me Anything**
  Jieyu Zhang, **Weikai Huang***, Zixian Ma*, Oscar Michel, Dong He, Tanmay Gupta, Wei-Chiu Ma, Ali Farhadi, Aniruddha Kembhavi, Ranjay Krishna
  NeurIPS 2024
- **m&m's: A Benchmark to Evaluate Tool-Use for multi-step multi-modal Tasks**
  Zixian Ma, **Weikai Huang**, Jieyu Zhang, Tanmay Gupta, Ranjay Krishna
  ECCV 2024

SKILLS

- **Languages and Tools**: Python, C++, Java, Docker, Bash, Git, LaTeX
- **DL Libraries**: PyTorch, Transformers, Huggingface Trainer, Peft, Accelerator, DeepSpeed, Flash-attn, Bitsandbytes, vLLM, veRL
- **3D/Vision Libraries**: Blender, Open3D, MeshLab, Detectron2, MMDetection, SAM, Depth Anything
- **Techniques**: Distributed training and model evaluation on clusters, Large scale data processing, Crowd worker data collection
- **Languages**: English, Chinese